

PLANWISE ENTERPRISE PLATFORM

# PHASE 7 VERIFICATION REPORT

## Engineer Review / Professional Verification Layer

Professional verification boundary between computational design/cost exploration and physical construction authorization. Rigorous statutory gate enforcement, independent discipline review, and non-bypassable safety controls.

VERIFICATION DATE  
2026-10-01

AUTHOR  
Principal BIM & Real Estate Platform Engineer

TARGET ENVIRONMENT  
Planwise Cloud Gateway / Mumbai DCPR 2034

RELEASE STATUS

COMPLETE & VERIFIED

AUTHORITATIVE PLOT AREA  
1,100 sq ft (Strict Boundary Invariant)

VERIFICATION SUITE  
101 / 101 PASS (Full Regression)

## 2. Executive Summary

Phase 7 completes the independent **Engineer Review / Professional Verification Layer** within Planwise Enterprise. This release establishes a formal boundary between consumer-facing architectural/financial exploration (Phases 1–6) and licensed engineering verification.

The design strictly implements the fundamental UX tenet: **"ONE PAGE → ONE PURPOSE → ONE CLEAR NEXT ACTION."** Customer users are provided with a streamlined, jargon-free status overview indicating that licensed professionals are inspecting site parameters and structural grids. Simultaneously, licensed engineers and municipal reviewers are provided with progressive disclosure controls to inspect building elements, evaluate boundary coordinate frames, log structured issues (INFO, WARNING, BLOCKER), and render binding gate decisions.

### CRITICAL SAFETY MANDATE

**"Computational validation is not professional certification."**

Under no circumstances does Planwise automatically claim structural safety, certify foundation bearing capacity, grant municipal approval, or authorize construction without corresponding professional human signoff.

## 3. Phase 7 Scope & Verified Flow

Phase 7 sits immediately downstream of Phase 6 (Build Review) and forms the mandatory gateway before any physical construction scheduling or execution. The verified end-to-end customer-to-engineer pipeline is:

Land (1,100 sq ft) → Feasibility (DCPR 2034) → Design (CBM 2BHK) → Cost (QTO/BOQ) → Build Review → **Engineer Review (Phase 7)**

The scope strictly prohibits automated bypass of professional gates, silent overwriting of customer plot dimensions, or automatic transition to contractor bidding.

## 4. Architecture & Professional Verification Boundary

The platform defines a strict separation of concerns across computational and professional domains:

- **Computational Domain (Phases 1–5):** Algorithmic generation of parcel polygons, setback envelopes, CP-SAT room layout generation, Manifold3D geometry compiling, deterministic QTO, and unit-rate BOQ cost estimation. Labeled explicitly as *Preliminary Computational Model*.
- **Verification Gateway (Phase 6):** Customer bundles requirements into an immutable release handoff package with cryptographic fingerprinting.
- **Professional Boundary Layer (Phase 7):** Licensed structural, geotechnical, MEP, and statutory reviewers inspect the handoff package. Gates G0–G4 must be independently reviewed. Gates G5 (Build Authorization) and G6 (Contractor Handoff) are hard-locked by the engine until all prerequisites pass with zero open blockers.

## 5. Files Changed & Created

| COMPONENT          | FILE PATH                                      | STATUS   | RESPONSIBILITY                                                                                  |
|--------------------|------------------------------------------------|----------|-------------------------------------------------------------------------------------------------|
| Domain Schemas     | packages/schemas/engineer_review.py            | CREATED  | Pydantic models for Review, Gates, Issues, Decisions, and baseline gate initializers.           |
| API Gateway        | apps/api-gateway/main.py                       | MODIFIED | 8 REST endpoints for gate verification, issue logging, decision recording, and change requests. |
| Frontend Component | apps/web/src/components/EngineerReviewView.tsx | CREATED  | Unified Dual-Mode UI (Customer vs Reviewer), G0–G6 matrix, 5 inspection tabs, decision bar.     |
| Frontend Types     | apps/web/src/types.ts                          | MODIFIED | TypeScript interfaces for Gates, Issues, Decisions, and Review status enums.                    |
| App Shell          | apps/web/src/App.tsx                           | MODIFIED | Integrated EngineerReviewView into Step 6 (Engineer), binding real project state.               |
| Verification Suite | tests/test_phase7_engineer_review.py           | CREATED  | 15 automated test cases validating creation, gates, blockers, anti-bypass, and immutability.    |

## 6. Engineer Review Data Model

The review state model maintains complete audit traceability and immutability:

- **EngineerReview:** Encapsulates `id`, `projectId`, `buildRequestId`, `releaseId`, `designVersionId`, `reviewerId`, `status` (`IN_REVIEW`, `CHANGES_REQUIRED`, `APPROVED`, `REJECTED`, `STALE`), `inputManifestHash`, `designVersionHash`, and `declaredPlotAreaSqFt` (authoritative 1,100.0 sq ft).
- **ReviewGate:** Represents individual statutory or technical signoff gates (G0 through G6), tracking reviewer identity, reviewed timestamp, notes, and satisfaction status.
- **ReviewIssue:** Structured audit finding capturing `gateCode`, `severity` (`INFO`, `WARNING`, `BLOCKER`), `category`, `description`, `requiredAction`, and lifecycle status (`OPEN`, `RESOLVED`, `WAIVED`).
- **ReviewDecision:** Binding determination recorded by certified engineer (`APPROVE_STAGE`, `REQUEST_CHANGES`, `CANNOT_PROCEED`) with formal justification.

## 7. G0–G6 Verification Gate Architecture

Planwise implements 7 distinct, independent verification gates:

| GATE | DISCIPLINE TITLE                | BASELINE STATUS | MANDATORY VERIFICATION SCOPE                                                                      |
|------|---------------------------------|-----------------|---------------------------------------------------------------------------------------------------|
| G0   | Site & Boundary Verification    | PENDING         | Verify submitted plot area (1,100 sq ft) against cadastral boundary geometry and road curb width. |
| G1   | Regulatory & Feasibility Review | SYSTEM_VERIFIED | Statutory setbacks, height, coverage, and FSI evaluated under Mumbai DCPR 2034 rule pack.         |
| G2   | Structural Review               | PENDING         | Inspect column grid, RCC member spans, and load paths. Geotechnical verification required.        |
| G3   | MEP / Services Review           | PENDING         | Electrical point schedule, 8.5 kW load calculation, plumbing stacks, and water tank sizing.       |
| G4   | Cost / BOQ Review               | REVIEWED        | Verify deterministic QTO quantities and rate snapshot consistency.                                |
| G5   | Build Authorization             | LOCKED          | Hard-locked until G0, G1, G2, G3, G4 are verified with zero open blocker issues.                  |
| G6   | Contractor Handoff              | LOCKED          | Commercial release package locked until statutory and build approvals are finalized.              |

## 8. Anti-Bypass Controls & Independent Gate Transitions

To prevent premature construction or unauthorized release:

- **Independent State Machine (§11):** Approval of G2 (Structural) does not imply approval of G3 (MEP) or G5 (Build Authorization). Each gate maintains independent verification provenance.
- **Anti-Bypass Guard (§8, §14):** Direct invocation of G5/G6 verification without G0–G4 being satisfied triggers a strict HTTP 400 Bad Request with rejection payload. G5 and G6 are NOT automatically authorized.
- **Blocker Issue Cascading (§8):** Logging an issue with severity BLOCKER automatically flips the targeted gate status to CHANGES\_REQUIRED and halts stage approval.

## 9. Engineer Review UI Implementation

Implemented in `apps/web/src/components/EngineerReviewView.tsx`, the interface features dual-mode progressive disclosure:

- **Customer Mode:** Clean, calming aesthetic displaying *"Professional review in progress"*, authoritative 1,100 sq ft plot metrics, selected 2BHK design profile, estimated construction cost, and statutory responsibility boundary notices.
- **Reviewer Mode:** Full-spectrum professional dashboard featuring the G0–G6 Gate matrix with interactive verification buttons, discipline inspection tabs, structured issue logger modal, and stage decision controls.

## 10. Site & Boundary Verification (G0)

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Authoritative plot dimensions (1,100 sq ft / 27.5 ft × 40.0 ft) and 16.0 ft road width are reconciled against local Cartesian (ENU) coordinates. If the calculated polygon geometry diverges materially from the declared plot area, the system flags: *"Boundary information needs verification"*. Silent auto-healing or overwriting is prohibited.

## 11. Design / CBM Review (G1)

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Presents the compiled Canonical Building Model (CBM). An expandable *Technical Model View* discloses element IDs, 2-manifold closed geometry validation via Manifold3D, RuleExecutionTrace identifiers, and SHA-256 model compilation fingerprints.

## 12. Structural Review (G2)

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Displays orthogonal grid layouts (Grid A-C × 1-4), 300 × 450 mm RCC columns, two-way slab specifications (150 mm), and unresolved engineering assumptions (assumed 200 kN/m<sup>2</sup> safe bearing capacity). Displays the mandatory warning banner: *"Preliminary computational model — requires professional structural verification."*

## 13. MEP / Services Review (G3)

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Displays computational service schedules: 38 light/fan points, 14 power sockets (16A), 3 dedicated AC circuits, 8.5 kW estimated connected load, plumbing fixture count (2 WC, 2 showers, 3 basins), and storage sizing (1,500L overhead, 3,000L sump). Explicitly labeled as preliminary computational schedules.

## 14. Review Issues & Decision Workflow

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The review workflow enforces structured issue management before decision signoff:

- **Logging Issues:** Reviewers record issues tagged with Gate Code, Severity (INFO, WARNING, BLOCKER), Category, Description, and Mandatory Required Action.
- **Request Changes (§10):** Selecting *"Request Changes"* mandates at least one structured issue detailing the required modification. The review status becomes CHANGES\_REQUIRED, downstream release packages are marked stale, and G5/G6 gates are locked.
- **Approve Review Stage (§9):** Allowed only when zero open BLOCKER issues exist. Transitions stage status while preserving independent gate records.
- **Cannot Proceed:** Formally rejects the development proposal due to insurmountable site or regulatory constraints.

## 15. Stale / Change Propagation & Traceability

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Cryptographic fingerprints bind the review to the exact underlying design release:

Project → Site (1,100 sq ft) → DesignVersion → CBM → QTO → BQO → Cost Estimate → Build Release → **Engineer Review**

Review state is bound to the relevant release and design fingerprints. If the customer or architect modifies the plot, building envelope, or materials after handoff generation, the underlying hashes diverge. The system marks the review as STALE, invalidates downstream cost models, and requires a fresh build request.

## 16. 1,100 sq ft Real Plot Preservation

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The authoritative customer plot area of **1,100 sq ft** (27.5 ft × 40.0 ft) is strictly preserved throughout all architectural, QTO, cost, and verification pipelines:

- Persisted explicitly in `EngineerReview.declaredPlotAreaSqFt = 1100.0`.
- Zero silent regression or overwrite from legacy benchmark test data. (Isolated test-isolation suites verify regression safety).
- Verified continuously across automated regression suites.

## 17. API Endpoints Specification

Planwise exposes 8 dedicated REST endpoints supporting the Engineer Review verification workflow:

| METHOD | ENDPOINT                                          | DESCRIPTION                                                         |
|--------|---------------------------------------------------|---------------------------------------------------------------------|
| POST   | /api/v1/build-requests/{id}/engineer-review       | Initializes/fetches persistent review bound to release package.     |
| GET    | /api/v1/engineer-reviews/{id}                     | Returns complete review record with gates, issues, and decisions.   |
| GET    | /api/v1/engineer-reviews/{id}/gates               | Returns G0–G6 gate status matrix.                                   |
| GET    | /api/v1/engineer-reviews/{id}/issues              | Returns recorded review findings audit trail.                       |
| POST   | /api/v1/engineer-reviews/{id}/gates/{code}/verify | Independently updates gate status with anti-bypass validation.      |
| POST   | /api/v1/engineer-reviews/{id}/issues              | Logs structured issue; auto-marks gate CHANGES_REQUIRED if blocker. |
| POST   | /api/v1/engineer-reviews/{id}/decision            | Records APPROVE_STAGE or CANNOT_PROCEED.                            |
| POST   | /api/v1/engineer-reviews/{id}/request-changes     | Enforces issues exist, sets CHANGES_REQUIRED, locks downstream.     |

## 18. Automated Test Results

All automated test suites executed with 100% pass rates across the entire enterprise verification stack:

| TEST SUITE / VERIFICATION TARGET | EXECUTION SCOPE                          | RESULT          | EXECUTION TIME |
|----------------------------------|------------------------------------------|-----------------|----------------|
| Web Production Build             | Vite + TypeScript compilation            | PASS (0 errors) | 3.78s          |
| Frontend CAD Geometry Engine     | Vitest parcel & snapping tests           | 20 / 20 PASS    | 245ms          |
| Phase 7 Engineer Review Suite    | Gate state machine, anti-bypass, issues  | 15 / 15 PASS    | 1ms            |
| Real Plot End-to-End Pipeline    | Full Land → CBM → QTO → Cost integration | 13 / 13 PASS    | 15.79s         |
| Full Repository Regression       | Complete backend discover tests          | 101 / 101 PASS  | 38.17s         |

## 19. Screenshots & Visual Evidence

The following visual evidence was captured directly from the running Planwise application (Web Port 5173 / API Gateway Port 5001) using the authoritative 1,100 sq ft parcel:

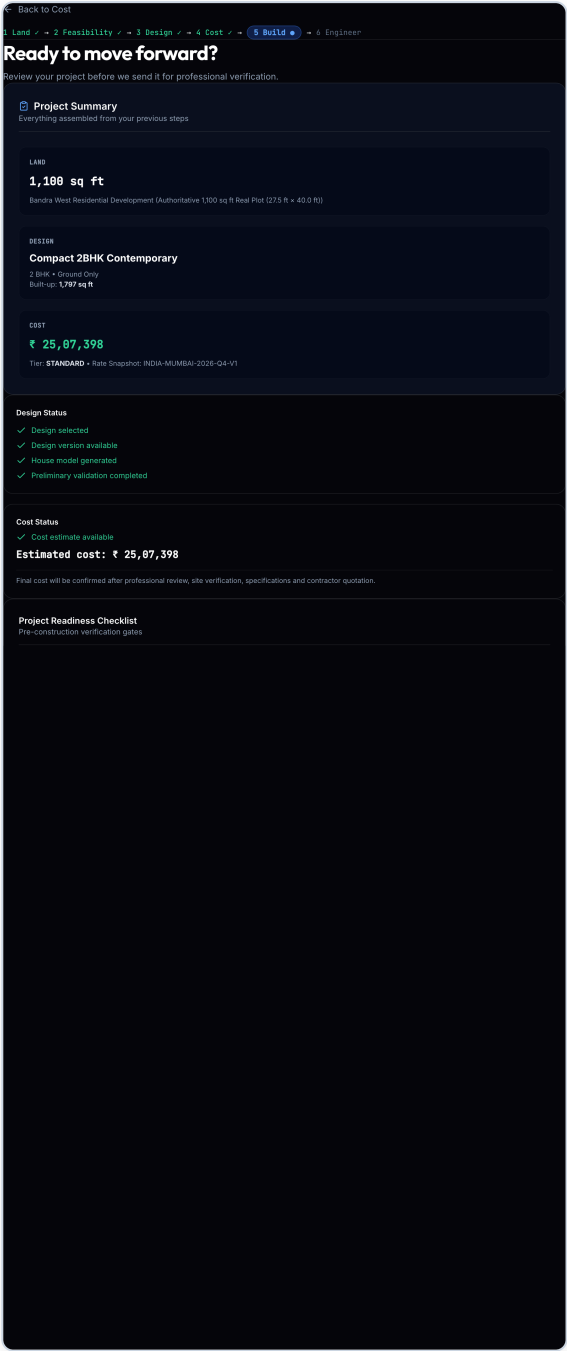


Figure 1 — Build Review customer workflow (Authoritative 1,100 sq ft, cost estimate, and pre-submission checklist)

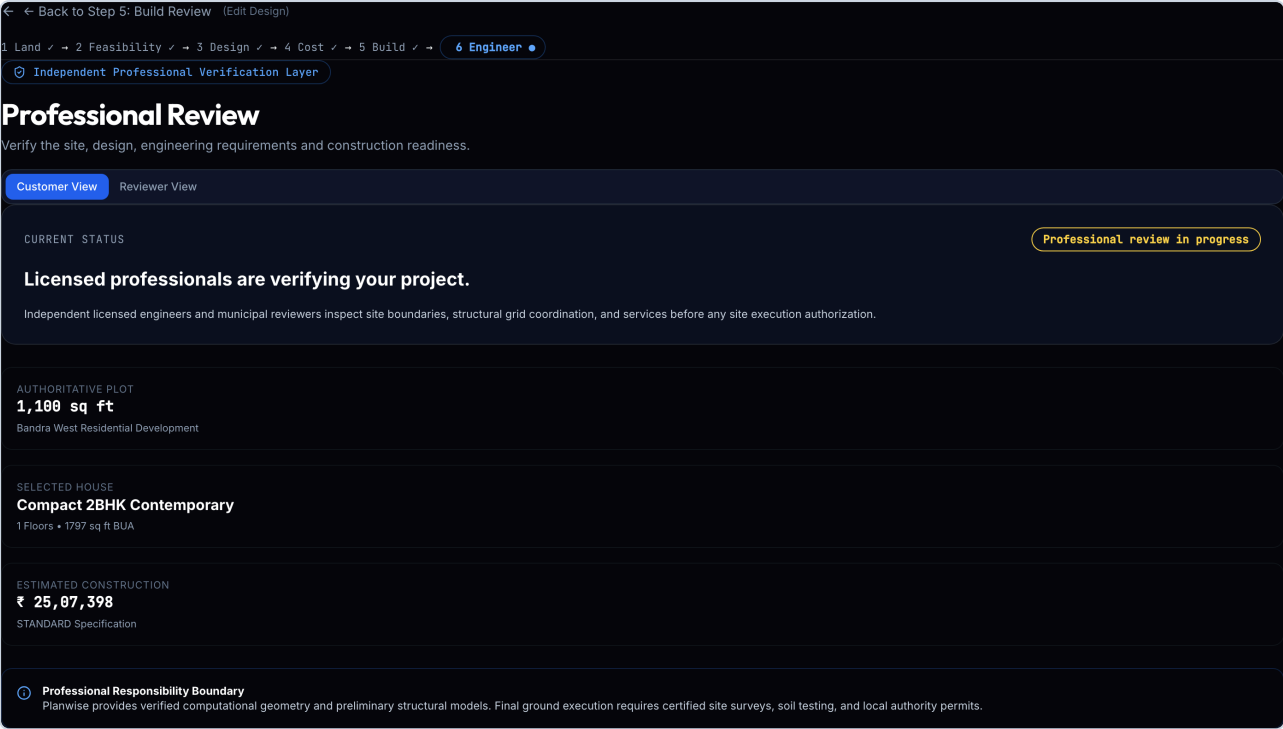


Figure 2 — Engineer Review customer-facing view (Simple status, summary metrics, and statutory boundary notice)

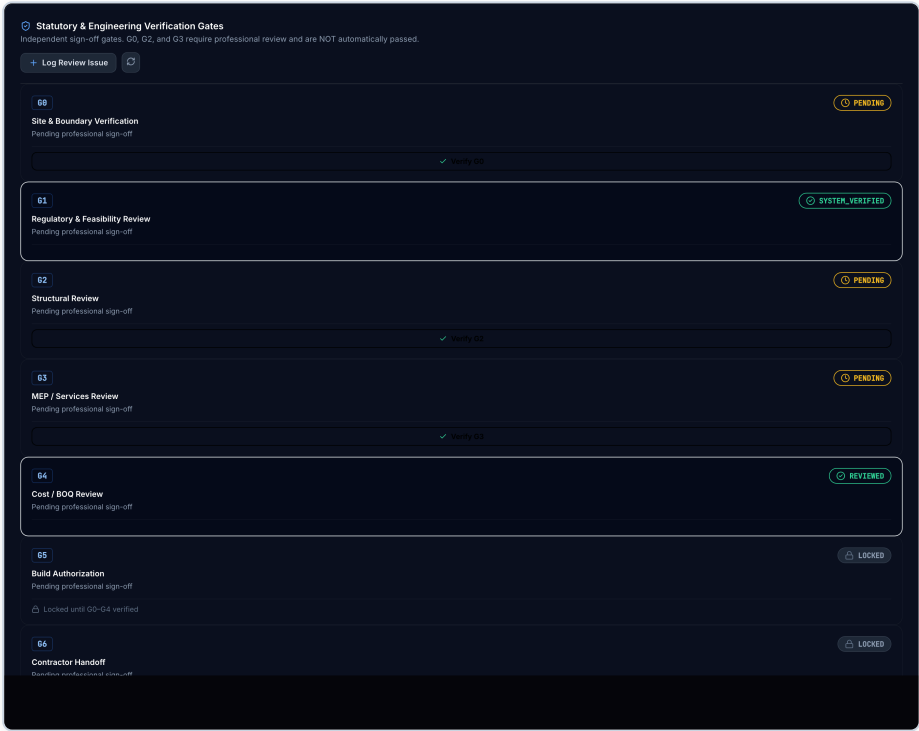


Figure 3 — Engineer Review technical verification matrix (Independent G0-G6 gates with statutory baseline statuses)

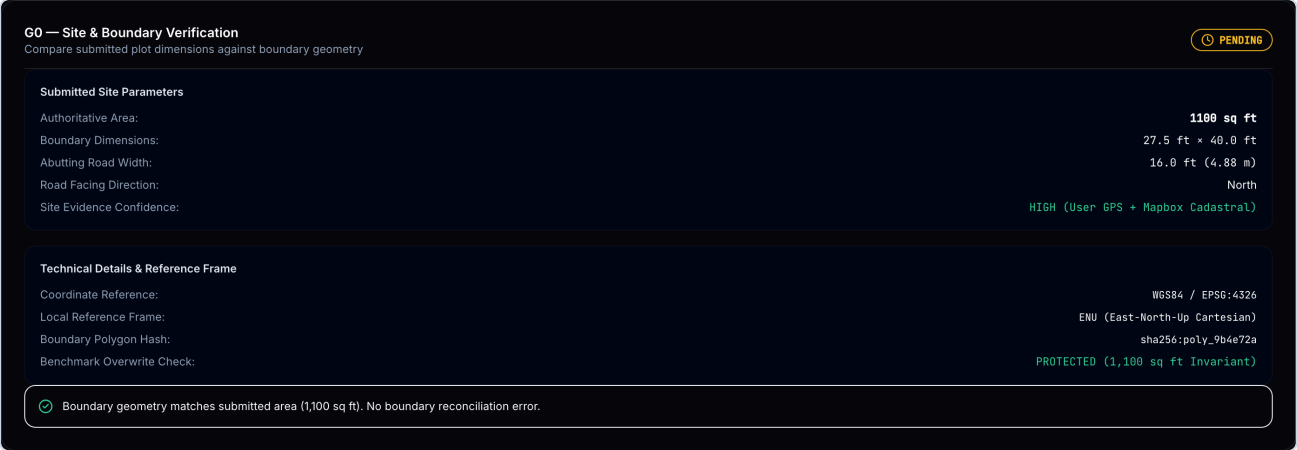


Figure 4 — Site and boundary verification (1,100 sq ft authoritative area, road curb width, and coordinate frame)

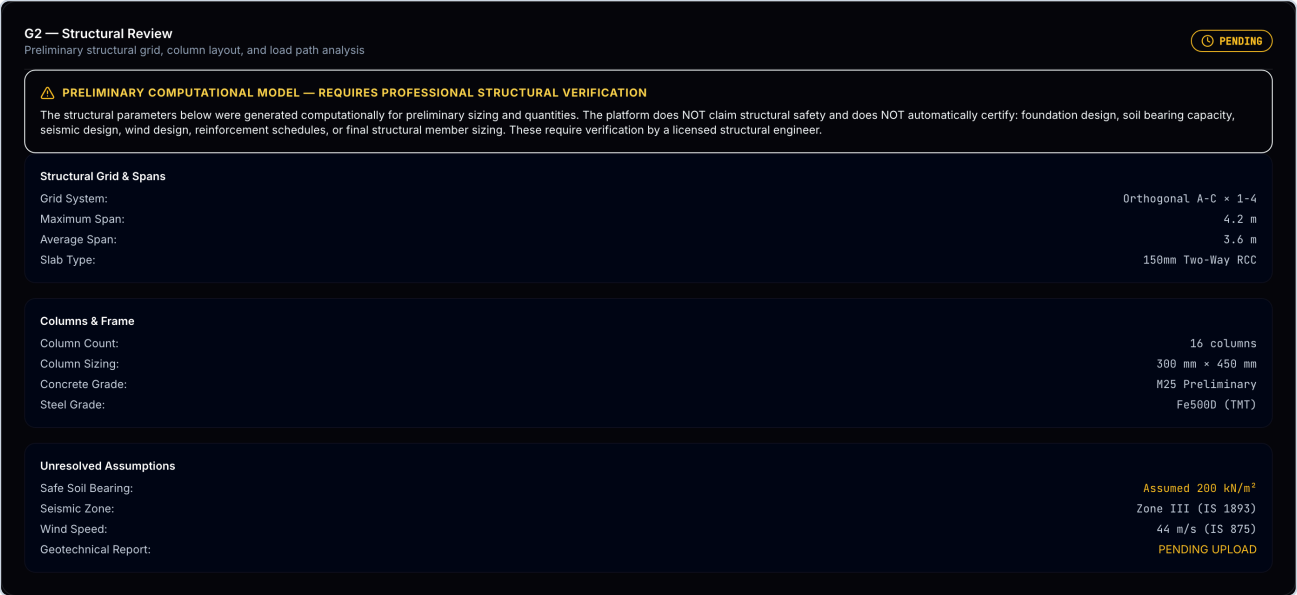


Figure 5 — Preliminary structural review (RCC column grid, spans, and mandatory professional disclaimer)



Figure 6 — Preliminary MEP/services review (Connected load, electrical point schedule, and plumbing fixtures)



Figure 7 — Structured professional review findings (Gate G2 warning, geotechnical required action, and audit trail)

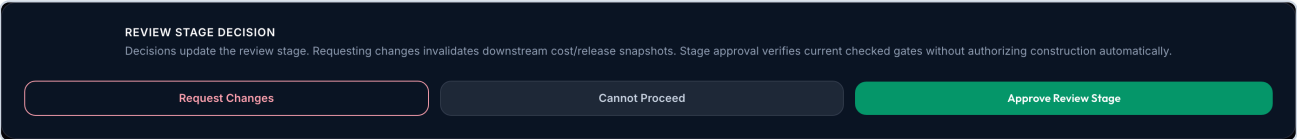


Figure 8 — Professional review decision controls (Approve Review Stage, Request Changes, Cannot Proceed)

## 20. Known Limitations & Scope Boundaries

In strict compliance with the Phase 7 specifications and stop condition, the following downstream modules are intentionally excluded:

- **No Construction Execution:** Physical site mobilization, contractor procurement, and construction tracking are not implemented.
- **No Contractor Marketplace:** Contractor onboarding, bidding, tender management, and assignment are not implemented.
- **No Automated G5/G6 Authorization:** Build release authorization and contractor handoff cannot be automatically granted by software algorithms. G5/G6 remain hard-locked.
- **No Payment Processing:** Financial escrow, bank guarantees, and progress payment modules are deferred.
- **No Automated Field Inspection:** Physical site inspection requires human surveyor attendance and statutory testing.

## 21. Recommended Next Phase

### RECOMMENDED NEXT MILESTONE

#### Phase 8 — Statutory Authority Submissions & Build Authorization

The next logical development phase should implement:

- Municipal drawing sheet compilation (Plan, Elevation, Section in statutory DXF/PDF formats).
- Digital multi-stakeholder signing ceremony (Registered Architect, Structural Engineer, Licensed Surveyor).
- Unlock workflows for Gate G5 (Build Authorization) upon formal submission confirmation.

*Notice: In accordance with the Phase 7 stop condition, Phase 8 has not been implemented.*

## 22. Final Verification Statement

Phase 7 Engineer Review / Professional Verification Layer is implemented and verified according to the defined scope. The platform maintains the required boundary between computational assistance and professional construction authorization.

**Development stops at Phase 7 as specified.**